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Research paper

Salidroside attenuates lipopolysaccharide (LPS) induced serum cytokines and depressive-like behavior in mice

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HIGHLIGHTS

- Salidroside increased BDNF.
- Salidroside decreased inflammation.
- Salidroside decreased depressive-like behavior.

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ABSTRACT

The aim of the study was to investigate the effects and possible underlying mechanism of salidroside (Sal) on lipopolysaccharide (LPS)-induced depression-like behavior in mice. Sal (12 mg/kg and 24 mg/kg) and fluoxetine (20 mg/kg) were administered intragastrically once daily for 5 days. At the 5th day, LPS (0.5 mg/kg) was injected intraperitoneally 30 min after drug administration. Levels of interleukin (IL)-6 and tumor necrosis factor (TNF)-α in serum were measured by ELISA. Levels of neurotransmitters like norepinephrine (NE) and 5-hydroxytryptamine (5-HT) in the prefrontal cortex were detected by HPLC-MS. Further, brain-derived neurotrophic factor (BDNF), tropomyosin-related kinase B (TrkB) and Nuclear factor-κB (NF-κB) in hippocampal was determined by western blot analysis. Our data showed that pretreatment with Sal dramatically attenuated LPS-induced inflammatory response, decrease of NE and 5-HT levels in the prefrontal cortex. In addition, Sal increased expression levels of BNDF and TrkB. These results suggested that Sal may play a neuroprotective role through the BDNF/TrkB signaling pathway.

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1. Introduction

Depression is a common mental disorder, psychiatric illness associated with morbidity and mortality. The pathological mechanism of depression is complicated and has not been fully elucidated. According to the 2012 World Health Organization (WHO) report, about 350 million people have suffered from depression which will become the second leading cause of disability worldwide by 2020 [25]. Current researches of antidepressants focus on monoamine neurotransmitter systems, including specific reuptake of neurotransmitters and their receptors [3]. However, the defects of antidepressants are low rates of response and severe adverse-effects. Recent researches in antidepressants are to discover new targets and to develop new therapeutics with higher efficacy and lower toxicity [2].

Several lines of evidence indicate that inflammation may play an important role in the development of depression [21]. The serum levels of pro-inflammatory cytokines, such as TNF-α, IL-6 and IL-1β, are elevated in the major depressive disorder (MDD) [12]. Associations between inflammatory markers and individual depressive symptoms have also been described. Lipopolysaccharides (LPS), which are found in the outer membrane of Gram-negative bacteria, could activate innate immune response and secrete pro-inflammatory cytokines. At the same time, LPS-induced depressive-like behaviors manifests 24 h after LPS administration. Antidepressants, including serotonin and norepinephrine reuptake inhibitors (SNRIs) and selective serotonin reuptake inhibitors (SSRIs), also have anti-inflammatory effects indicating that anti-inflammatory effects can be used as replacement ther-

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apy for LPS-induced depression [1,17]. A growing number of herbal medicines have been found to be effective in the treatment of psychiatric diseases. Salidroside, which is extracted from *Rhodiola rosea* L., has long been used as an adaptogen in traditional Tibetan medicine. It possesses comprehensive pharmacological properties such as anti-aging, anti-cancer, anti-inflammation and hepatoprotective, anti-oxidative and immunological enhancement. For example, salidroside has been found to be remarkably protective against myocardial ischemia and reperfusion injury [16]. Nevertheless, no report has been issued on anti-depression effects of salidroside, traditional herbs may become a novel pharmacotherapy in the treatment of depression. Therefore, our study is to investigate whether salidroside is effective in LPS-induced depressive-like behavior.

It is widely acknowledged that the level of 5-HT and NE in the brain plays an important role in the pathogenesis of depression. Some researches indicate that pro-inflammatory cytokines have a direct impact on the metabolism of 5-HT and NE [26]. Inflammation inducers (like LPS) could activate indoleamine-2,3-dioxigenase (IDO) enzyme and up-regulate 5-HT transporter (SERT) in the brain. IDO enables the conversion of tryptophan to kynurenine, the low serum of tryptophan may confine the tryptophan effect, and finally limited 5-HT synthesis in the brain. The SERT is responsible for the reuptake of 5-HT, that is to say, up-regulation of SERT may lead to decrease of 5-HT level in the brain [23]. As well as pro-inflammatory cytokines can influence the synthesis and reuptake of NE, leading to the reduction of NE levels in the brain. These cytokine-induced changes in neurotransmitter function correlated with the development of depression.

Accumulating evidence suggest that neurotrophins play critical roles in neuron survival, plasticity, neurogenesis and synaptogenesis. As an important member of the neurotropic family, brain-derived neurotrophic factor (BDNF) is most widely expressed. The "neurotrophin hypothesis of depression" holds that BDNF is an important factor in the pathophysiology of depression [15]. The down-regulation of BDNF is involved in the pathophysiology of depression and antidepressants can restore the BDNF level. Recently, the involvement of BDNF and its receptor TrkB in antidepressant effects have been intensively investigated [6]. The antidepressant treatment is able to attenuate the degradation of BDNF found in brain and blood samples of depressed patients [7,14]. Furthermore, transgenic mice with reduced BDNF-mediated signaling in brain are insensitive to some antidepressants [30]. We explored whether BDNF and TrkB activation is involved in the neuroprotective effect of salidroside in the present study.

2. Materials and methods

2.1. Animals

Total 60 male ICR mice that were 8 weeks of age (18–22 g) were purchased from comparative medicine center of Yangzhou University, and adapted one week before experiments in a standard condition with controlled temperature at 24 ± 1 °C, relative humidity of 50% and 12 h light/12 h dark cycle. All experiments and animal care were performed by the Animal Research Committee of China Pharmaceutical University.

2.2. Drugs and assay kits

Salidroside (Sal, provided by The Second Military Medical University, shanghai, China; purity ≥99%). Fluoxetine (FLU), purchased from Changzhou Siyao Pharmaceuticals Co., Ltd. (Changzhou, China). Lipopolysaccharides (LPS, *Escherichia coli* serotype 0111: B4, No. L-2630, Sigma–Aldrich, St. Louis, MO, USA), ELISA kits (TNF-α, IL-6; R&D Systems, Inc., Minneapolis, MN).

2.3. Experimental protocol

Sixty animals were randomly divided into five groups ($n = 12$): (1) control group, (2) vehicle group: LPS (0.5 mg/kg, i.p.) (3) positive group: fluoxetine (20 mg/kg, i.g.) (4), (5) treatment of salidroside(12 and 24 mg/kg, i.g.). All drugs were administered intragastrically once daily for 5 days. At the 5th day, LPS (0.5 mg/kg) was injected intraperitoneally 30 min after drug administration. Blood samples from eye socket were collected 90 min after LPS injection and then it was centrifuged at 3500 rpm for 10 min, and the serum were collected and stored at −80 °C until the assessment of serum interleukin (IL)-6 and tumor necrosis factor (TNF)-α levels by ELISA. Behavioral tests were measured with 24 h after LPS injection. After behavioral tests, all the mice were sacrificed, the prefrontal cortex was rapidly separated from the brain, all the prefrontal cortex were stored at −80 °C for analyses of 5-HT and NE levels by HPLC-MS, and brain-derived neurotrophic factor (BDNF) in hippocampal was determined by western blot.

2.4. Behavioral evaluations

2.4.1. Tail suspension test (TST)

The TST was conducted based on previously reported methods [32]. Mice were individually suspended upside down by their tails, mice were suspended for 6 min, and the immobility time was measured during the last 4 min. In this study, observers who were blind to the treatment group independently.

2.4.2. Forced swimming test (FST)

The FST was conducted as previously described [29], with minor changes. In brief, individual mice were placed into an acrylic cylinder (height: 25 cm, diameter: 15 cm) containing 10 cm of water (depth) at 22–25 °C, The time of immobility was defined as the absence of active, escape-oriented behaviors. Mice was forced to swim for 6 min, and the immobility time was measured during the finally 4 min. In this study, the observers were blind to the group of the mice.

2.5. Measurement of cytokines

Blood samples were drawn from orbit, then it was centrifuged at 3500 rpm for 10 min, and the serum was collected. The concentrations of cytokine TNF-α and IL-6 in the serum were measured by ELISA kit, according to the instructions of manufacturer.

2.6. Measurement of 5-HT and NE levels

Levels of 5-HT and NE were measured by HPLC coupled with MS. A Waters ACQUITY™ triple-quadrupole tandem mass spectrometer (Water Corp., Milford, MA, USA) was connected to the UPLC system via electrospray ionization (ESI) interface. UPLC was performed on a Waters ACQUITY UPLC system (Waters, Milford, MA, USA), consisting of a binary solvent delivery system, an online degasser, an auto-sampler and a photo-diode array detector (PDA) system. An ACQUITY UPLC™ BEH C_{18} (2.1 × 100 mm I.D., 1.7 μm, Waters, Milford, USA) column was used for all the analyses. The mobile phase composed of A (0.1% formic acid, v/v) and B (acetonitrile) with a gradient elution: 0–5 min, 90–70% A; 5–7 min, 70–0% A; 7–9 min, 0% A; 10 min, 90%. The flow rate of the mobile phase was 0.4 ml/min and the column, auto-sampler temperature was maintained at 30 and 10 °C, respectively. High purity nitrogen was used as the nebulizer and auxiliary gas, argon was as the collision gas. ESI source parameters for NE and 5-HT as follows: capillary = 1.00 kV,

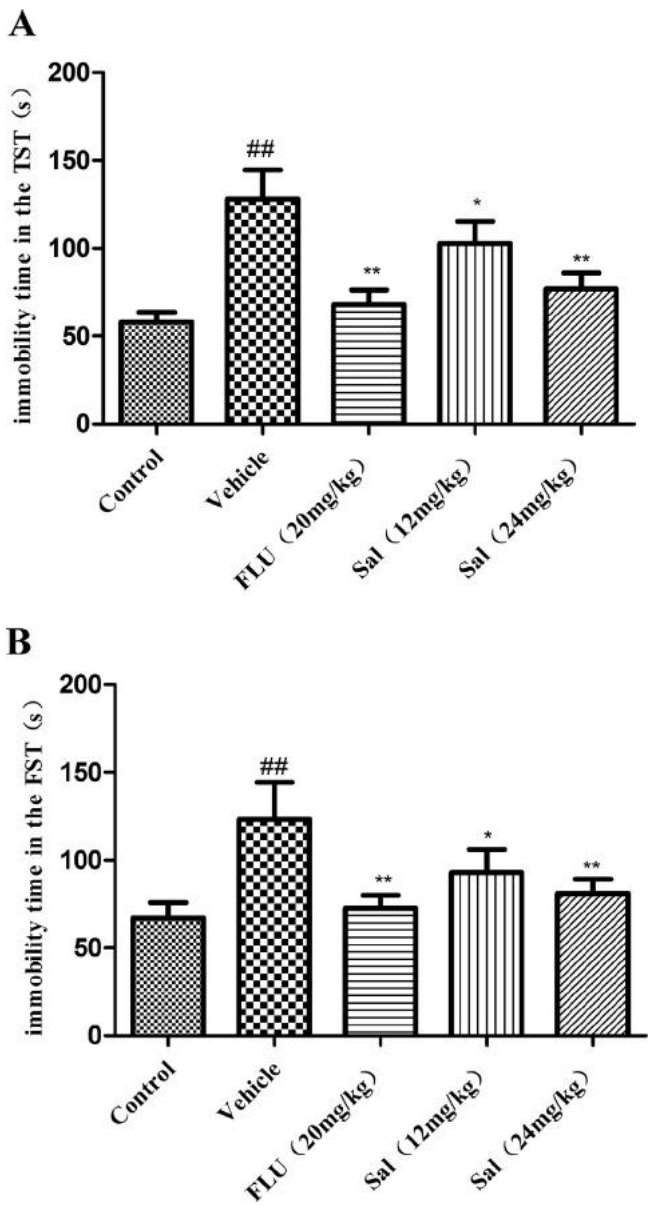

Fig. 1. Effects of salidroside (Sal) on immobility time in the TST(A) and FST(B) in LPS-induced mice. Behavioral tests were performed 24 h after LPS administration. All values are expressed as means ± SDs. Compared with control group: $^{\#}P < 0.05$, $^{\#\#}P < 0.01$; Compared with vehicle group: $^{*}P < 0.05$, $^{**}P < 0.01$.

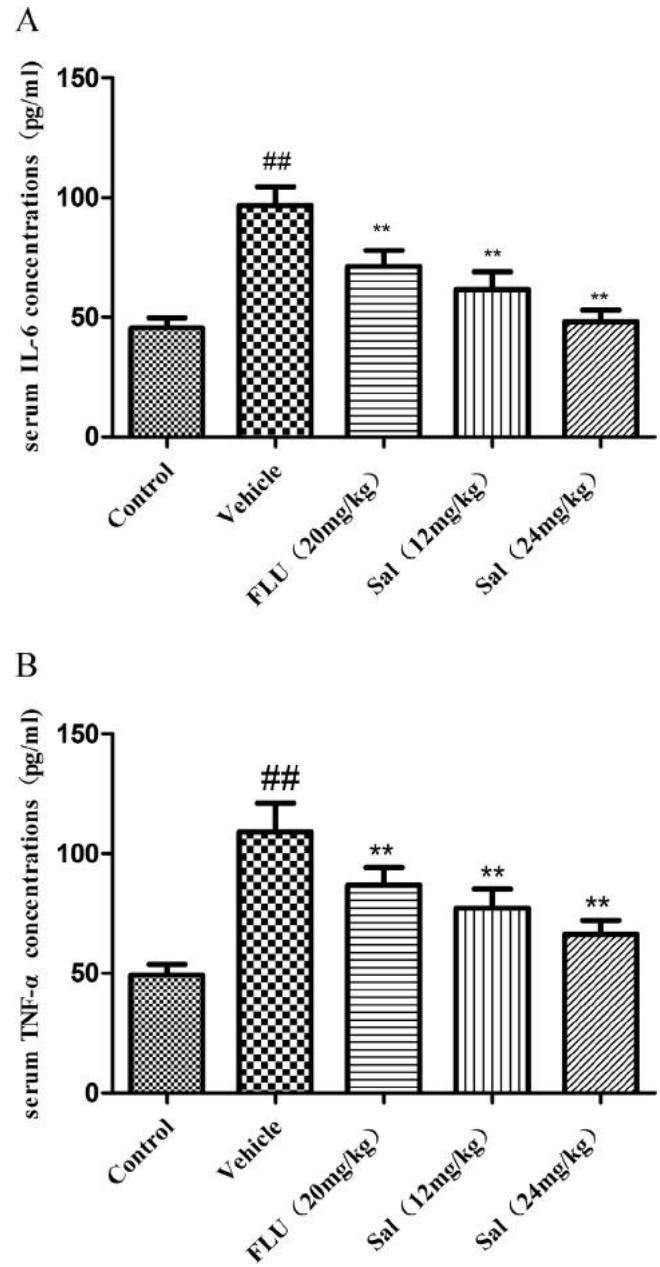

Fig. 2. Effects of Salidroside (Sal) on plasma IL-6 (A) and TNF-α (B) levels in LPS-induced mice. Blood sample was collected 90 min after LPS administration. All values are expressed as means ± SDs. Compared with control group: $^{\#}P < 0.05$, $^{\#\#}P < 0.01$; Compared with vehicle group: $^{*}P < 0.05$, $^{**}P < 0.01$.

extractor = 4.00 V, cone = 30.00 V. The temperatures of the source and desolvation were set at 120 °C and 350 °C, respectively, and nitrogen was used as the desolvation gas (600 L/h) and cone gas (120 L/h). For collision-induced dissociation (CID), argon was used as the collision gas at a flow rate of 0.17 ml/min (approximately 3.96×10^{-3} mbar).

2.7. Western blotting of BDNF, TrkB and NF-κB

The brain tissues were homogenized, washed with normal saline, and incubated in buffer in addition to a protease inhibitor cocktail (Sigma, St. Louis, MO) to obtain extracts of brain proteins. The samples were loaded to 12% SDS-PAGE gels and were electro-transferred to nitrocellulose. The blots were incubated with the appropriate concentration of specific antibody (anti-BDNF, anti-TrkB and anti-NF-κB, CST). After washing, the blots were incubated with horseradish peroxidase-conjugated second antibody. The membranes were stripped and reblotted with anti-β-actin antibody (Sigma) to verify the equal loading of protein in each lane.

2.8. Statistical analysis

All data in the figures and tables were presented as the mean ± standard deviation (SD). One-Way ANOVA was used to compare among multiple groups and P value was analysis by Dunnett test. All data were processed with SPSS13.0 software. $P < 0.05$ was considered to be statistically significant.

3. Results

3.1. Effects of salidroside on behavioral evaluations

3.1.1. Effects on immobility time in TST

The data of TST is shown in Fig. 1A. LPS administration significantly increased immobility time in the TST compared to the control group ($P < 0.01$). Whereas Sal significantly decreased immobility time in the TST compared with the vehicle group.

3.1.2. Effects on immobility time in FST

The data of FST is shown in Fig. 1B. LPS administration exhibited remarkable increases of immobility time in the FST when compared to the control group ($P < 0.01$), while pretreatment with Sal notably shortened immobility time in the FST compared to the vehicle group.

3.2. Effects of salidroside on serum cytokine levels

As shown in Fig. 2A and B. LPS administration exhibited significantly increased the serum levels of IL-6 ($P < 0.01$) and TNF-α ($P < 0.01$) when compared to the control group. However, Sal significantly attenuated the serum levels of IL-6 ($P < 0.01$) and TNF-α ($P < 0.01$) compared with the vehicle group.

3.3. Effects of salidroside on 5-HT and NE levels in LPS-induced mice

As shown in Fig. 3A and B, LPS administration significantly decreased 5-HT ($P < 0.01$) and NE ($P < 0.01$) levels in the prefrontal cortex ($P < 0.01$) in contrast with the control group. As expected, Sal significantly resulted in significantly increased 5-HT ($P < 0.01$) and NE ($P < 0.01$) levels in the prefrontal cortex compared with the vehicle group.

3.4. Effects of salidroside on protein of BDNF, TrkB and NF-κB in LPS-induced mice

We examined BDNF, TrkB and NF-κB protein in LPS-induced mice. As shown in Fig. 4, We found that LPS administration significantly down-regulated BDNF, TrkB and induced the expression of NF-κB compared with the control group ($P < 0.01$, $P < 0.05$). In contrast, Fluoxetine (20 mg/kg) and Sal (12, 24 mg/kg) significantly increased the protein levels of BDNF and TrkB. Meanwhile, they both showed to down-regulate NF-κB compared with the vehicle group ($P < 0.01$, $P < 0.05$).

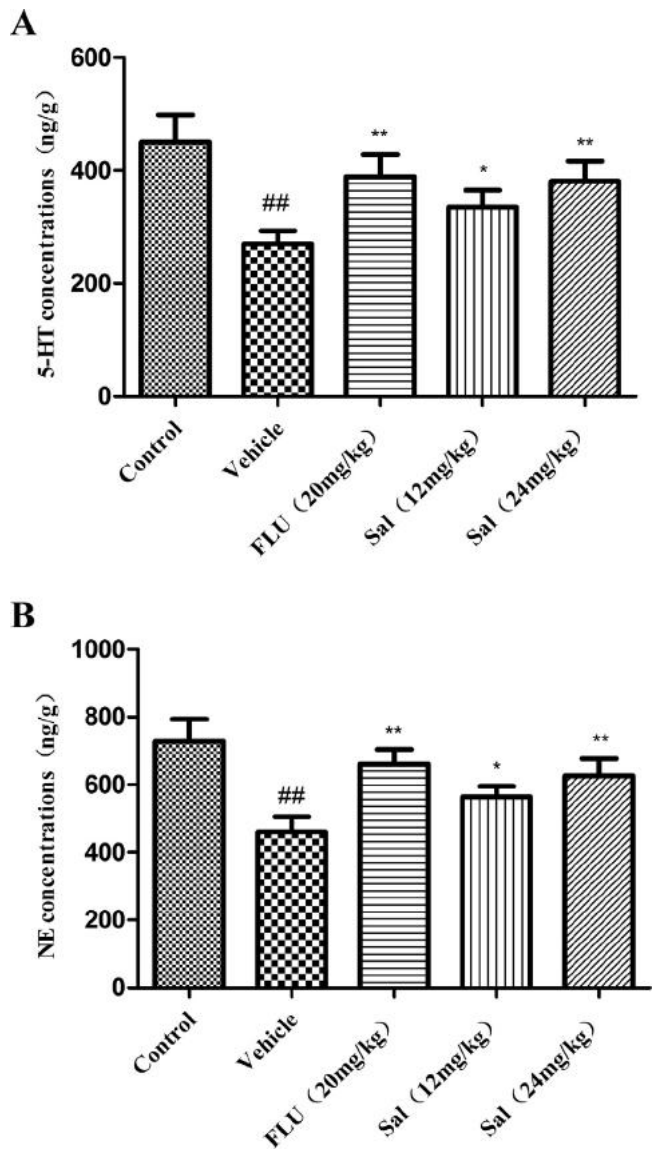

Fig. 3. Effects of Salidroside (Sal) on 5-HT (A) and NE (B) levels in LPS-induced mice. The prefrontal cortex sample was collected after measuring behavioral tests. All values are expressed as means ± SDs. Compared with control group: $^{\#}P < 0.05$, $^{\#\#}P < 0.01$; Compared with vehicle group: $^{*}P < 0.05$, $^{**}P < 0.01$.

4. Discussion

The present study is the first to demonstrate that the salidroside may play an important role in anti-depressive effects. The results of our study showed that LPS administration could induce depressive-like behavior in mice, which are consistent with previous studies [9,28]. In our study, LPS administration leads to dramatic increases of IL-6 and TNF-α in the serum. However, pretreatment with salidroside for 5 days could ameliorate depression-like behaviors and attenuate the inflammatory cytokines and NF-κB protein induced by LPS injection. Besides, our study also showed that the anti-depressive effects of Sal might be related to the metabolism of 5-HT and NE, as well as up-regulation of BDNF and TrkB expressions in the brain. To conclude, the present of our study revealed that sal improved depression due to its significant effects on monoaminergic responses, anti-inflammatory function and regulating BDNF/TrkB signaling pathway.

Salidroside presents pharmacological properties such as anti-inflammatory [31] and immunological enhancement activities [11]. Nevertheless, the development of depressive disorders is accompanied by activation of immune-inflammatory pathways consisting of innate immune system and inflammatory process [10,18]. These findings suggest that inflammation played an vital role in depressive symptoms, and inhibition of inflammatory process might relieved these symptoms. In previous studies, LPS induced inflammation is associated with depressive-like behavior in mice, such as forced swimming test (FST) and tail suspension test (TST) [13]. In our study, the result showed that Sal could significantly reduce the immobility time in FST and TST, which indicated that Sal could attenuate LPS-induced depressive-like behavior.

Antidepressants drugs, such as fluoxetine, the previous reported that it possesses anti-inflammation in the periphery, fluoxetine could attenuated LPS-induced increase of TNF-α level [28]. At the same time, IL-6 was elevated in the serum of MDD patients [5]. Disrupted sleep–wake cycles and sleep deprivation lead to activation of NF-κB [24], a transcription factor with a key role in triggering through the control of the expression of a series of inflammatory genes, including IL-6 and TNF-α, Indicating antidepressant action

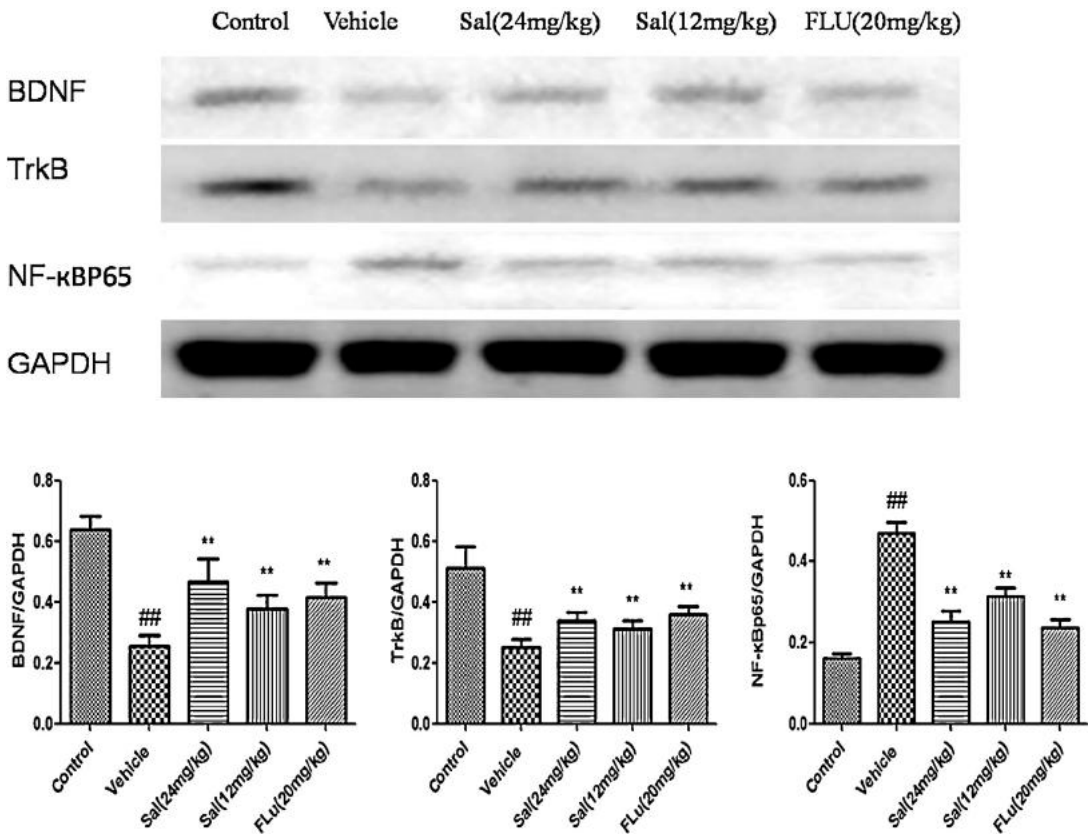

Fig. 4. Effects of Salidrosid (Sal) on hippocampal BDNF, TrkB and NF-κB in LPS-induced mice. Relative optical density of BDNF, TrkB and NF-κB expression quantified via Western blot. All values are expressed as means ± SDs. Compared with control group: $^{\#}P < 0.05$, $^{\#\#}P < 0.01$; Compared with vehicle group: $^{*}P < 0.05$, $^{**}P < 0.01$.

was related to anti-inflammatory effects. Our study demonstrated that the anti-depressive effect of Sal might be closely related to its anti-inflammatory effect indicated by decreases of TNF-α, IL-6 levels and depression of NF-κB in the brains of LPS-induced mice.

Accumulating evidences have revealed that BDNF, the most abundant neurotrophin in the brain, is involved in the pathophysiology and treatment of depression [6]. Local infusion of BDNF into the brain has antidepressant-like effects in behavioral animal models of depression. In this present study, pretreatment of Sal prevented the down-regulation of BDNF. The BDNF-mediated regulation might be a part of their activated pathways to provide neuroprotection [33]. BDNF exerts its neuroprotective functions through its specific high-affinity receptor tyrosine kinase B (TrkB), the binding of BDNF will lead to the phosphorylation of TrkB, then activate the downstream molecules and initiate the signaling event. It has been reported that BDNF binds to TrkB receptors and activates PI3K/Akt signaling, resulting in inhibition GSK-3β phosphorylation that has an antidepressant effect [19]. some study also indicated that it can activate MAPK/ERK pathway [8], the MAPK cascade is mediated by activation of the transcription factor cAMP response element-binding protein (CREB). An increased expression of CREB in the hippocampus produces an antidepressant-like effect. Furthermore, several lines of evidence indicate that CREB activation is also able to stimulate the expression of BDNF. Our result suggested that salidroside exhibited its neuroprotective effect through the BDNF/TrkB signaling pathway, as evidenced by up-regulation of BDNF and TrkB in the brain.

Previous studies demonstrated that depression is strongly associated with impairments in 5-HT, NE, and DA neurotransmission, and these monoaminergic systems are thought to be vital targets for the treatment of depression [22,27]. 5-HT and NE has been believed to play an important role in the pathophysiology of depression and the main mechanism of depressants [20]. In addition, the metabolism of 5-HT and NE were related to pro-inflammatory cytokine on pathways [4]. In this study, the result showed that pretreatment of Sal could effectively reverse the levels of 5-HT and NE in LPS-induced mice, which indicated that antidepressant efficiency of Sal might relate to alter the monoaminergic responses and further study should be need.

5. Conclusions

In conclusion, we report that salidroside has anti-depressive effects in the LPS-induced mice of depression, and the anti-depressive effects might be associated with the alterations of 5-HT and NE, reduction of inflammation effects including inflammatory cytokines and NF-κB expression, and activation of BDNF and TrkB.

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